GRIDINTEL · UTILITY INTELLIGENCE

DisCo operations

ATC&C and its parts, revenue not realised, collection by customer class, reliability and hours of supply, assets requiring attention, outages and alarms.

SAMPLE: SYNTHETIC DATA

This report is a sample. It shows the format and the methods GridIntel would deliver; it is not a report of real results. Every figure in it is calculated from a designed demonstration dataset, and none describes a real network, site, customer or organization.

A designed demonstration dataset for September 2026. It describes no real network, customer, meter reading, outage, bill or payment.

FOR
DisCo management

RECORDS OF
Savanna Electricity Distribution (synthetic)

PERIOD
14 September 2026

REPORT ID
GI-OPS-2026-09-14-BFD2EC4E

FIGURES COMPUTED
20 Oct 2026, 10:11 WAT, from data as of 15 Sep 2026, 00:00 WAT

DEFINITION
Version 1

DATA FINGERPRINT (SHA-256)
4b772d3711bfe1a55a6bf221904d2401a9f0b36f39f0d3412a5bbc6d8ef745f1

IN BRIEF

SAIDI
8.1 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIFI
1.5 • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CAIDI
5.4 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIDI ATTRIBUTED TO THE DISTRIBUTION NETWORK
0.0 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

The loss account and the money are not given for a day: A loss account and collection efficiency compare energy delivered, energy billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is delivered, and payments arrive on days of their own, so for a single day these are not on one basis. They are given in the month's report.

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1 Losses: ATC&C and its parts

How much of the energy the DisCo received was not turned into money, and in which part of the chain it was lost.

Not available for a day. A loss account and collection efficiency compare energy delivered, energy billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is delivered, and payments arrive on days of their own, so for a single day these are not on one basis. They are given in the month's report.

2 Revenue not realised

What the commercial and the collection losses are worth, in money, and on which feeders.

Not available for a day. A loss account and collection efficiency compare energy delivered, energy billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is delivered, and payments arrive on days of their own, so for a single day these are not on one basis. They are given in the month's report.

3 Billing and collection by customer class

Who was billed, who paid, and who left the most uncollected.

Not available for a day. A loss account and collection efficiency compare energy delivered, energy billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is delivered, and payments arrive on days of their own, so for a single day these are not on one basis. They are given in the month's report.

4 Reliability and hours of supply

How often, and for how long, customers were without supply, and whether each feeder gave the hours its band commits to.

SAIDI
8.1 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIFI
1.5 • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CAIDI
5.4 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIDI ATTRIBUTED TO THE DISTRIBUTION NETWORK
0.0 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

Hours of supply against the band, by feeder

Each day is tested against the band minimum on its own; the average over the period is shown beside it.

	BAND	BAND MINIMUM	AVERAGE SUPPLIED	DAYS MET	DAYS BELOW
Market Road 11 kV feeder	Band A	20.0 h/day • C	21.0 h/day • C	1 • C	0 • C
Old Town 11 kV feeder	Band C	12.0 h/day • C	15.0 h/day • C	1 • C	0 • C
Farm Road 11 kV feeder	Band D	8.0 h/day • C	12.0 h/day • C	1 • C	0 • C
Government Avenue 11 kV feeder	Band B	16.0 h/day • C	15.0 h/day • C	0 • C	1 • C

Method: GridIntel reference hours of supply (gridintel.supply_hours.reference 0.2.0).

> The indices by substation and feeder, where interruptions began and the DisCo's reported reliability figures are in the Reliability and service standards report for the same period.

5 Assets requiring attention

Which assets had a source alarm, a derived condition or an interruption on them, each fact with its own origin.

> **Why an asset is listed.** An asset is listed when a source alarm is standing on it or on its monitor at the as-of time, or GridIntel derived a condition for it or its monitor in the period, or one or more interruptions attributed to the distribution network began at it in the period. Order: most standing alarms first; then conditions that hold at the as-of time; then most readings above rating; then most interruptions begun; then by id. It is an ordering of facts, not a score.

The assets

In the order the rule gives. Loading is apparent power against rating, from telemetry; alarms are as source systems recorded them; conditions are GridIntel's, by rule.

	CLASS	PEAK LOADING	READINGS ABOVE RATING	SOURCE ALARMS STANDING	DERIVED CONDITIONS	INTERRUPTIONS THAT BEGAN HERE
Government Avenue transformer 3 Hillcrest 33/11 kV injection substation · Government Avenue 11 kV feeder	Distribution transformer	104.1% • C	5 • C	None	Loaded above rating; no source alarm raised	0 • C
Riverbank transformer Riverside 33/11 kV injection substation · Old Town 11 kV feeder	Distribution transformer	112.8% • C	3 • C	None	Loaded above rating; no source alarm raised	0 • C

Method: GridIntel reference equipment loading (gridintel.loading.reference 0.1.0).

Maintenance records: Not available. No source supplies work orders, inspections or repairs, so none is shown and no asset is called maintained or overdue.

Age and condition: Not available. The registry holds no commissioning date, test result or condition assessment for transformers or feeders, so no asset is ranked by age or health.

6 Outages and alarms at the end of the period

What was wrong when the period closed: interruptions still in progress, alarms standing, and conditions holding.

Interruptions in progress

Outages whose customers, or some of them, had not got supply back at the end of the period, largest first.

> None: every interruption in the outage log had been restored by the end of the period.

Restoration not recorded

Outages with no restoration time whose source does not say they are open: a gap in the record, not a finding that supply is still off.

> None.

Source alarms standing

As source systems recorded them, most severe first. Accounts behind an alarm are those connected there, not a count of customers without supply.

> None was standing at the end of the period.

1 alarm(s) have no recorded raise time, so whether they were standing cannot be told.

0 alarm(s) raised were cleared by the end of the period.

Derived conditions

Found by GridIntel from telemetry under fixed rules, for the period. They are calculated, and are not alarms from any source system.

	RULE	FIGURE	AT THE END OF THE PERIOD	SOURCE ALARM
Riverbank transformer Transformer	Loaded above rating	112.8% • C	Not holding	No source alarm raised
Government Avenue transformer 3 Transformer	Loaded above rating	104.1% • C	Not holding	No source alarm raised

Method: GridIntel reference equipment loading (gridintel.loading.reference 0.1.0).

Whether a monitor has gone quiet cannot be told: the source's record of device check-ins is not complete, so a missing check-in may only be a missing record.

Loaded above rating: Loading (apparent power ÷ rated capacity, or highest phase current ÷ rated current) was above 100% of rating at one or more telemetry readings in the period.

Monitor quiet: The device's last check-in is more than 120 minutes before the as-of time, or no check-in from it is held in the 24 hours before it. Applied only where the record of check-ins is complete over those hours.

7 Method, sources and how to reproduce this report

Where every figure came from, and how a reader can check that this document is the one the records give.

Methodologies

METHODOLOGY	VERSION	WHAT IT SAYS OF ITSELF
GridIntel reference equipment loading `gridintel.loading.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.
GridIntel reference reliability indices `gridintel.reliability.reference`	`0.3.0`	GridIntel reference calculation — not a regulatory figure.
GridIntel reference hours of supply `gridintel.supply_hours.reference`	`0.2.0`	GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Sources

- Synthetic alarm list (synthetic)
- Synthetic billing and vending (synthetic)
- Synthetic meter data (synthetic)
- Synthetic outage log (synthetic)
- Synthetic asset registry (synthetic)
- Synthetic telemetry (synthetic)

> Every source above is synthetic: designed demonstration data, not records of a real organization.

How to reproduce it

1. Build the report with definition version 1, for 14 September 2026, from the Savanna Electricity Distribution (synthetic) records.
2. The data fingerprint must be `4b772d3711bfe1a55a6bf221904d2401a9f0b36f39f0d3412a5bbc6d8ef745f1`. SHA-256 over the sorted SHA-256 digests of every distinct result the report's services were given by the repositories, each written as JSON with its keys in order.
3. With the same definition, methodologies and fingerprint the report ID is the same: GI-OPS-2026-09-14-BFD2EC4E. A different ID means something above differs.
4. The figures with their status, origin, unit and method are in the CSV of the same report ID: gridintel-operations-2026-09-14-SAMPLE-SYNTHETIC.csv.

8 Regulatory references

> How far this report carries the fields of the instruments below, as read from the instruments themselves. It is not a claim that the report conforms to any regulator's or programme's template: none has been checked against the template it would be filed on.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	7–8. Distribution losses; technical losses: energy delivered to the DisCo, energy billed, physical losses; monthly	**PARTLY** Section 1 gives the energy account and ATC&C with its parts, by DisCo and feeder, for a month. The technical loss is an estimate from a loss study, not metered, and is labelled as one.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	15–16. Billing efficiency; collection efficiency: energy billed and received, revenue billed and collected; monthly	**CARRIED** Sections 1 and 3, for a month. Collection is on a cash basis: money received in the period against bills issued in it.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	1–3. SAIDI, SAIFI, CAIDI; monthly	**PARTLY** Section 4 gives the DisCo's indices in brief; by substation and feeder they are in the Reliability and service standards report.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	12–14, 17–20. Staff cost, productivity, unit cost, naira per kWh, outstanding debt, capacity costs	**NOT CARRIED** Not held: GridIntel holds no financial accounts of the DisCo.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	s. 9. Monthly customer complaints report and register	**NOT CARRIED** Not held: GridIntel holds no complaints records.

9 Glossary and key

ATC&C	Aggregate technical, commercial and collection loss: the share of energy received that was not paid for. Technical + commercial + collection loss.
Technical loss	Energy lost in the network itself. Here an estimate from a loss study, not a measurement.
Commercial loss	Energy delivered but not billed: a residual, delivered less billed, which carries the uncertainty of the technical-loss estimate.
Collection loss	The share of energy billed whose bills were not paid in the period.
Billing efficiency	Energy billed ÷ energy received.
Collection efficiency	Revenue collected in the period ÷ revenue billed in the period, on a cash basis.
Revenue not realised	An estimate of revenue lost to commercial and collection loss in the period, each valued at the rate where it occurs. Not annualised.
Not available for a day	A loss account and collection efficiency need energy, billing and collection over one span; a single day does not give one, so they are given for a month only.
SAIDI	System Average Interruption Duration Index: the hours of interruption the average customer served had in the period. Customer-hours interrupted ÷ customers served.
SAIFI	System Average Interruption Frequency Index: the interruptions the average customer served had in the period. Customer interruptions ÷ customers served.
CAIDI	Customer Average Interruption Duration Index: the average length of an interruption, in hours. SAIDI ÷ SAIFI.
Network-attributable	The part of an index caused by interruptions that began on the distribution network (33 kV and 11 kV lines, distribution transformers, the low-voltage network), under the reference methodology.
Service band	NERC's Service-Based Tariff band of a feeder, with its minimum average hours of supply a day: A 20, B 16, C 12, D 8, E 4.
Provisional	A figure that includes an interruption still open at the end of the period, counted to its end. It is calculated, not estimated, and is final only when the restoration is recorded.
Source alarm	An alarm as a source system (SCADA, a relay, a monitor) recorded it. Observed; GridIntel does not decide whether it is true.
Derived condition	Something GridIntel worked out from telemetry under a fixed rule (loaded above rating, monitor quiet). Calculated; never shown as an alarm.
Accounts behind	Active customer accounts connected behind an asset, from the registry. It says who is connected there, not who is without supply.
Restoration not recorded	An outage with no restoration time whose source does not say it is open: a gap in the record, not a finding that supply is still off.
Measured	Read from a meter, a device or a source record.
Reported	A figure published by someone else, shown as they stated it. It is never altered.
Calculated	Computed by GridIntel from records, under a named and versioned methodology.
Estimated	An estimate, not a measurement; the method is named with it.
Derived	A residual: what is left when other figures are subtracted. It carries their uncertainty.
Insufficient data	An input the figure needs is missing. No value is given rather than a guess, and never a zero.
Not available	No source holds this figure, or the data does not cover the period.
Same basis	A reported figure and a calculation count the same things over the same period by the same rule. Only figures on the same basis are compared.
Data fingerprint	A SHA-256 digest of every record the report was built from. The same records give the same fingerprint; any change to them gives another.

Report ID	The report's code, its period and the first characters of a digest of its definition, methodologies and data fingerprint. The same report of the same data has the same ID.

The marks beside a figure

• **OK**	Every input is present and measured.
≈ **Estimated inputs**	Calculated; every input is present, but at least one was estimated rather than measured.
! **Insufficient data**	An input this figure needs is missing. No value is shown rather than a guess.
∅ **Not computable**	The inputs exist but the calculation is undefined for them.
– **Not available**	The platform has no source for this figure.
M **Measured**	Read from a meter, device or source record.
R **Reported**	Published by someone else; shown as stated.
C **Calculated**	Computed by GridIntel under a named methodology.
E **Estimated**	An estimate, not a measurement.
D **Derived**	A residual: what is left after subtracting other figures. It inherits their uncertainty.

SAMPLE: SYNTHETIC DATA. End of report GI-OPS-2026-09-14-BFD2EC4E.